

Asymmetric – Symmetric cryptosystem

Module III

Symmetric Versus Asymmetric key cryptography

Characteristic	Symmetric key cryptography	Asymmetric key cryptography
Key used for encryption/decryption	Same key is used for encryption and decryption	One key used for encryption and another, different key is used for decryption
Speed of encryption/decryption	Very fast	slower
Size of resulting encrypted text	Usually same as or less than the original clear text size	More than the original clear text size
Key agreement/exchange	A big problem	No problem at all
Number of keys required as compared to the number of participants in the message exchange	Equals about the square of the number of participants, so scalability is an Issue.	Same as the number of participants, so scales up quite well
Usage	Mainly for encryption and decryption (confidentiality), cannot be used for digital signatures (Integrity and non-repudiation)	Can be used for encryption and decryption (confidentiality) as well as for digital signatures (Integrity and non-repudiation checks)

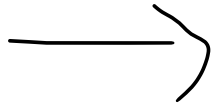
Key Exchange Algorithm

Public key of Agent X is A
Private key of Agent X is d
Symmetric key is K

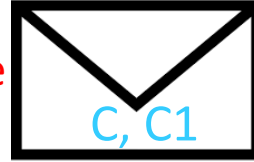
Encrypt K using
public key cryptosystem
 $E_B(K) = C1$

Encrypt Plaintext X using
Symmetric key cryptosystem
 $E_K(M) = C$

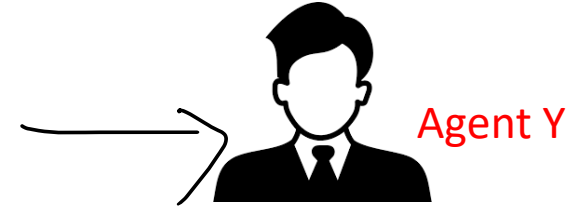
Agent X



Digital envelope



Public key of Agent Y is B
Private key of Agent Y is e



Decrypt using Asymmetric key
Cryptosystem $D_e(C1) = K$

Decrypt using symmetric key
Cryptosystem $D_K(C) = M$

Here, both sender and receive
Need to know about
symmetric and asymmetric algorithms used.

public area